Assignment

Subject: programing_fundamental_lab_3

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Section: D

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Task1

Assume that the following variables are defined:

Int age;

double pay;

char section;

write a single cin statement that will read input into each of these variables.

```
#include<iostream>

using namespace std;

int main()

{

        int age;

        double pay;

        char section;

        cout<<"enter your age pay and section: "<<endl;

        cin>>age>>pay>>section;

        return 0;

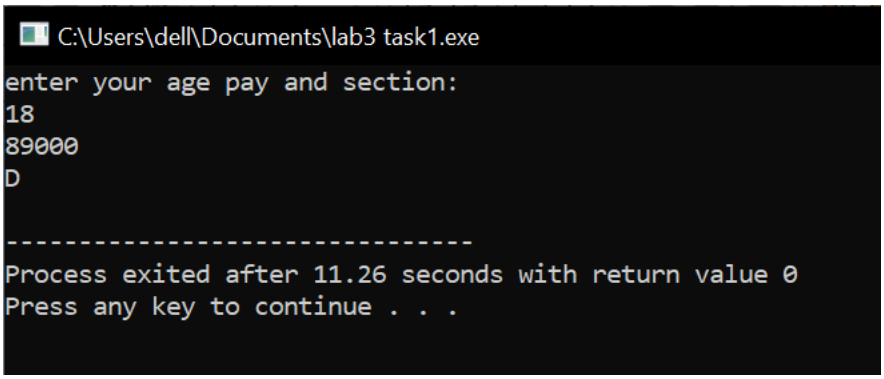
}
```

Output:

Enter your age pay and section: 18

89000

D



Task2

Complete the following table by writing the values of each expression in the value column according c++ language rules.

Expression	value
28/4-2	5
6+12*2-8	22
4+8*2	20
6+17%3-2	7
2+22*(9-7)	46
(8+7)*2	30
(16+7)%2-1	0
12/(10-6)	3
(19-3)*(2+2)/4	16

Task3

Assume a program has the following variable definitions:

int units;

float mass;

double weight;

weight=mass*units;

Which automatic data type conversion will take place?

A. mass is demoted to an int, units remains an int, and the result of mass * units is an int.

B. units is promoted to a float, mass remains a float, and the result of mass * units is a float.

C. units is promoted to a float, mass remains a float, and the result of mass * units is a double.

```
#include<iostream>

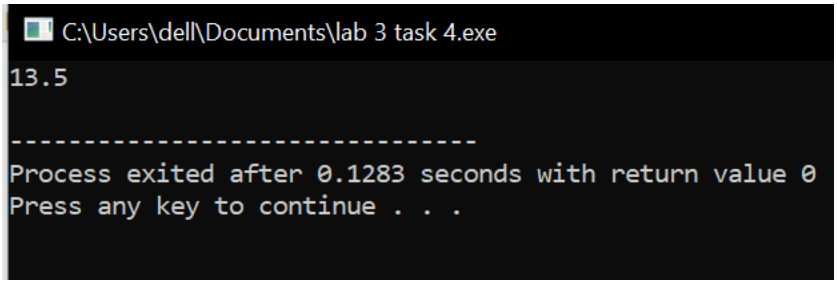
using namespace std;

int main()
```

```
{
        int units=3;
        float mass=4.5;
        double weight;
        weight = mass*units;
        cout<<weight<<endl;
        return 0;
}
```

Output:

13.5



Write answer is B.

B. units is promoted to a float, mass remains a float, and the result of mass * units is a float.

Task4

Assume a program has the following variable definition:

int a,b=2;

float c=4.2;

and the following statement: a=b*c;

what value will be stored in a?

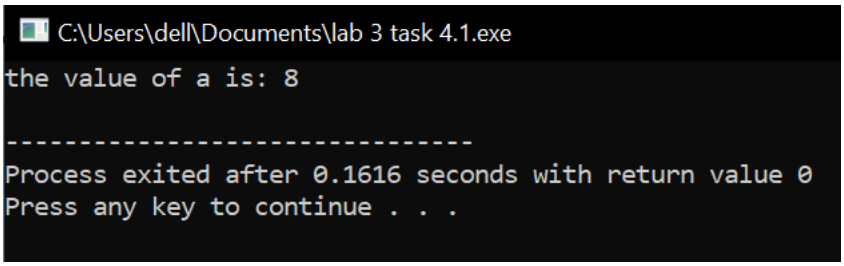
A. 8.4
B. 8
C. 0
D. None of the above

```
#include<iostream>
using namespace std;
int main()
{
```

```
        int a,b=2;
        float c=4.2;
        a = b * c;
        cout<<"the value of a is: "<<a<<endl;
        return 0;
    }
```

Output:

the value of a is: 8



Answer is B.

B. 8

Task5

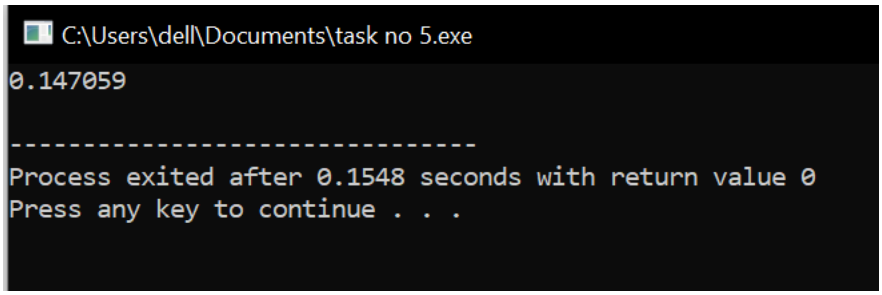
assume that qty and salesReps are both integers. use a type cast expression to rewrite the following statement so it will no longer perform integer division.

unitsEach = qty / salesReps;

```
#include<iostream>
using namespace std;
int main()
{
        int salesReps=34,qty=5;
        double z =(double) qty / salesReps;
        cout<<z<<endl;
        return 0;
}
```

Output:

0.147059



Home Tasks

Task1

Each of the following programs has some errors. Locate as many as you can.

Program 1.

```
using namespace std;

void main()

{

double number1, number2,sum;

cout << "Enter a number:";

Cin <<number1;

cout << "Enter another number:";

cin <<number2;

number1+number2=sum;

cout”The sum of the two numbers is “<<sum

}
```

Errors:

1.There is we can’t include any header file in the start.

2.In the 6th and 8th line we use << these special symbols instead of >>.

3.In the 10th line there is missing << and also missing semi colons at the end.

Program 2:

```
#include <iostream>

using namespace std;

void main()

{

int number1,number2;

float quotient;

cout << "Enter two numbers and I will divide\n";

cout << "the first by the second for you.\n";

cin>>number1,number2;

quotient=float<static_cast>(number1)/number2;

cout<<quotient

}
```

Errors:

1.we use int main() instead of void main().

2.cin>> number1,number2; read only number1. Use cin>>number1>>number2;

3.The cast syntax is incorrect. Use quotient=static_cast<float>(number1)/number2;

3.The final cout<< statement need semi colons at the end.

4.It's good idea to check that number2 is not the zero for this we use if function.

Task2

Average of Values to get the average of a series of values, you add the values up and thendivide the sum by the number of values. Write a program that stores the following values infive different variables: 28, 32, 37, 24, and 33. The program should first calculate the sum ofthese five variables and store the result in a separate variable named sum. Then, theprogram should divide the sum variable by 5 to get the average. Display the average on thescreen.

```
#include<iostream>

using namespace std;

int main()

{

        int a=28;

        int b=32;

        int c=37;

        int d=24;

        int e=33;

        int sum;

        double average;

        sum =a+b+c+d+e;

        cout<<"the sum is: "<<sum<<endl;

        double avarage= static_cast<double>(sum)/5;

        cout<<"the average is: "<<avarage<<endl;

        return 0;


}
```

Output:

the sum is: 154

the average is: 30.8

